

An explicit symmetric Keller counterexample in six variables and a 44-variable vanishing witness

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with heavy assistance from ChatGPT 5.6 Sol

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Abstract

Starting from the recently announced three-dimensional noninjective Keller map, we give two explicit consequences. Meng's classical gradient lift yields an identity-linear six-variable gradient Keller map with symmetric Jacobian and an exact three-point fiber. A rank-compressed homogenization of a 13-variable stable model gives a 22-variable cubic homogeneous noninjective Keller map. The de Bondt–van den Essen symmetric reduction then gives a 44-variable homogeneous quartic Hessian-nilpotent polynomial with 538 monomials and an exact collision. Expanded potentials and collisions are supplied as machine-readable exact certificates.

Verification disclaimer. The author is a complete amateur and cannot independently verify these claims. This is an experiment in the limits of AI-assisted mathematics, not an established result. Every argument, attribution, and novelty claim requires independent expert review. Exact checks are evidence, not peer review.

1 A six-variable symmetric Keller map

Let $i^2 = -1$. The identity-linear normalization of the announced map [2] is

$$\begin{aligned}\Phi_1 &= x - \frac{3}{2}x^2y - \frac{1}{2}x^3z, \\ \Phi_2 &= y + 3x(1 + xy)^2z + 3xy^2(4 + 3xy), \\ \Phi_3 &= (1 + xy)^3z + y^2(1 + xy)(4 + 3xy).\end{aligned}\tag{1}$$

It satisfies $\det J\Phi = 1$ and maps

$$p_0 = (0, 0, -\frac{1}{4}), \quad p_1 = (1, -\frac{3}{2}, \frac{13}{2}), \quad p_2 = (-1, \frac{3}{2}, \frac{13}{2})\tag{2}$$

to $q = (0, 0, -\frac{1}{4})$. For $A, B \in \mathbb{C}^3$, put

$$X = A + iB, \quad \Lambda = \frac{A - iB}{2}, \quad \boxed{\mathcal{S}(A, B) = \Lambda^T \Phi(X)}.\tag{3}$$

Theorem 1. *The gradient map $\nabla \mathcal{S} : \mathbb{C}^6 \rightarrow \mathbb{C}^6$ has symmetric Jacobian, identity linear part, and Jacobian determinant one. The three distinct points*

$$z_j = (p_j/2, -ip_j/2) \in \mathbb{Q}(i)^6\tag{4}$$

all map to $(q/2, -iq/2) = (0, 0, -1/8, 0, 0, i/8)$. The potential $\mathcal{S}$ has degree eight and 204 monomials when expanded.

Proof. For $f(X, \Lambda) = \Lambda^T \Phi(X)$,

$$\text{Hess}(f) = \begin{pmatrix} \sum_k \Lambda_k \text{Hess}(\Phi_k) & J\Phi^T \\ J\Phi & 0 \end{pmatrix}. \quad (5)$$

Swapping the two blocks of three columns makes this matrix block upper triangular, so $\det \text{Hess}(f) = (-1)^3 (\det J\Phi)^2 = -1$. Equation (3) is the pullback of f by

$$C = \begin{pmatrix} I & iI \\ \frac{1}{2}I & -\frac{i}{2}I \end{pmatrix}, \quad \det C = i. \quad (6)$$

Hessians transform by congruence, giving $\det \text{Hess}(\mathcal{S}) = (\det C)^2 (-1) = 1$. Since $J\Phi(0) = I$, direct multiplication gives $\text{Hess}(\mathcal{S})(0) = I_6$. Finally, $\nabla f(p_j, 0) = (0, q)$; applying C^T proves the collision. $\square$

This is Meng's classical gradient lift [4, 5], specialized to the new map and normalized over $\mathbb{Q}(i)$.

2 Rank-compressed homogenization

Lemma 2. *Let*

$$\Psi(X) = X + H_2(X) + H_3(X) : \mathbb{C}^n \rightarrow \mathbb{C}^n, \quad \det J\Psi = 1, \quad (7)$$

where H_d is homogeneous of degree d . Suppose $H_3 = BK$ for a constant $n \times r$ matrix B and a cubic homogeneous $K : \mathbb{C}^n \rightarrow \mathbb{C}^r$. Define

$$h(X, U, t) = (tH_2(X) + t^2BU, -K(X), 0). \quad (8)$$

Then h is cubic homogeneous and Jh is nilpotent. If $p \neq q$ and $\Psi(p) = \Psi(q)$, then $(p, K(p), 1)$ and $(q, K(q), 1)$ collide under $W \mapsto W + h(W)$.

Proof. For an indeterminate s , a Schur complement in $I + sJh$ gives

$$\begin{aligned} \det(I + sJh) &= \det(I + stJH_2 + s^2t^2BJK) \\ &= \det J\Psi(stX) = 1. \end{aligned} \quad (9)$$

Thus the characteristic polynomial of Jh is T^{n+r+1} . At $t = 1$,

$$(X, K(X), 1) + h(X, K(X), 1) = (\Psi(X), 0, 1), \quad (10)$$

which proves the collision statement. $\square$

The standard homogenization [1] takes $r = n$. Lemma 2 retains only the constant coefficient span of the cubic components.

3 The 22-variable cubic map

The factor-reusing stable reduction in [9] produces

$$\Psi = X + H_2 + H_3 : \mathbb{C}^{13} \rightarrow \mathbb{C}^{13}, \quad \det J\Psi = 1, \quad (11)$$

in the variable order

$$X = (x, y, z, a_1, b_1, a_2, b_2, a_3, b_3, a_4, b_4, a_5, b_6). \quad (12)$$

The six triangular degree-reduction operations and final output shear are reproduced in the human-readable source accompanying this PDF. They lift the collision (2) to

$$\begin{aligned} \pi_0 &= (0, 0, -\frac{1}{4}, 0, 0, 0, \frac{1}{2}, 0, 0, 0, 0, 0, 0), \\ \pi_1 &= (1, -\frac{3}{2}, \frac{13}{2}, -1, -2, \frac{9}{2}, -10, \frac{3}{2}, \frac{3}{4}, -\frac{9}{4}, -6, -\frac{27}{8}, \frac{9}{2}). \end{aligned} \quad (13)$$

Exact coefficient row reduction shows that the thirteen components of H_3 span an eight-dimensional space. Take $K = (K_1, \dots, K_8)$ with

$$\begin{aligned} K_1 &= \frac{1}{2}x(a_1z + b_1x), \\ K_2 &= a_2a_3z - 3a_2y^2 + 3b_2a_3x + b_3xy + 12xy^2, \\ K_3 &= -b_1a_3a_4 + a_3b_4y - 3a_4xz + a_5xz - b_6xy + 3xyz, \\ K_4 &= 3x^2y, \\ K_5 &= 2b_1a_3a_4 - 2a_3b_4y + 6a_4xz - 2a_5xz + 2b_6xy - 5xyz, \\ K_6 &= xy^2, \\ K_7 &= -b_1xy - 7a_2a_3z + 21a_2y^2 - 21b_2a_3x - a_3xz - 7b_3xy - 84xy^2, \\ K_8 &= a_4xy. \end{aligned} \quad (14)$$

Then $H_3 = BK$, where

$$B(U_1, \dots, U_8) = (U_1, U_2, U_3, 0, -3U_2, U_4, U_5, 0, 0, U_6, U_7, U_8, 0). \quad (15)$$

The verifier checks (15) coefficient by coefficient and checks $\text{rank } B = 8$. Lemma 2 therefore gives a 22-variable map with 72 cubic monomials across 21 active nonlinear coordinates. Its collision points are $(\pi_j, K(\pi_j), 1)$, where

$$K(\pi_0) = 0, \quad K(\pi_1) = (-\frac{17}{4}, -\frac{45}{8}, \frac{99}{16}, -\frac{9}{2}, -\frac{177}{8}, \frac{9}{4}, \frac{213}{8}, \frac{27}{8}). \quad (16)$$

4 The 44-variable quartic

For the 22-variable map h , define

$$\boxed{\mathcal{P}(A, B) = \mathbf{i} \sum_{j=1}^{22} h_j(A + \mathbf{i}B) B_j.} \quad (17)$$

The de Bondt–van den Essen characteristic-polynomial identity [3] implies that $\text{Hess}(\mathcal{P})$ is nilpotent exactly when Jh is nilpotent. Hence

$$\Gamma(Z) = Z - \nabla \mathcal{P}(Z) : \mathbb{C}^{44} \rightarrow \mathbb{C}^{44} \quad (18)$$

has Jacobian determinant one.

The cubic collision transports explicitly. If r_0, r_1 collide under $x \mapsto x + h(x)$, let $M_j = I + Jh(r_j)^T$, set

$$y_1 = 0, \quad y_0 = M_0^{-1}(-\mathbf{i}h(r_1) + \mathbf{i}h(r_0)), \quad (19)$$

and define $z_j = (r_j - \mathbf{i}y_j, y_j)$. Direct differentiation gives $\Gamma(z_0) = \Gamma(z_1)$. The two 44-tuples are in `output/collision.json`. Their coordinates are in $\mathbb{Q}(\mathbf{i})$, with numerator at most 261 in absolute value and denominator at most 16.

Theorem 3. *The polynomial $\mathcal{P}$ is a homogeneous quartic in 44 variables with 538 monomials and nilpotent Hessian. The map $Z - \nabla\mathcal{P}(Z)$ is noninjective. Moreover,*

$$\Delta^m \mathcal{P}^m = 0 \quad (m \geq 1), \quad \Delta^m \mathcal{P}^{m+1} \neq 0 \quad (20)$$

for infinitely many m .

Proof. Only the last assertion remains. Zhao [6] proved that Hessian nilpotence is equivalent to $\Delta^m \mathcal{P}^m = 0$ for all $m \geq 1$, and that the formal inverse of $Z - t\nabla\mathcal{P}$ is governed by

$$Q_t = \sum_{m \geq 0} \frac{t^m}{2^m m! (m+1)!} \Delta^m \mathcal{P}^{m+1}. \quad (21)$$

Eventual vanishing would make the inverse polynomial, contradicting the explicit collision. $\square$

5 Certificates, scope, and priority

The expanded files and SHA-256 hashes are

file	SHA-256
symmetric_potential_sparse.json	1e0c97e1c4965c3e...498015c3c3f5fdd4
symmetric_collision.json	6b5b546f24e839a1...89936a711add0
potential_sparse.json	2a91272816188884...2a65e0751789f4
collision.json	aeab7adb021c07de...89936a711add0

The full hashes are in MANIFEST.md. The symbolic verifier checks the Hessian congruence, the rank-eight factorization, identity (9), exact nilpotence guards, and both collisions. A dependency-free checker reads only the JSON, differentiates both sparse potentials, and verifies the six-variable identity linear part and both finite fibers.

The six-variable result is an explicit instance of Meng’s known reduction, not a new reduction theorem. Zhang [7] left the Vanishing Conjecture consequence existential. Our earlier note [9] gave a 54-variable certificate. Thompson [8] posted a 24-variable cubic homogeneous reduction at 03:29:42 UTC on 21 July 2026. His work contains the same rank-compression principle in executed form and has priority for that idea. The narrower quantitative contribution here is the different $13 + 8 + 1 = 22$ construction and its 44-variable quartic.

Searches of arXiv, MathOverflow, and public GitHub code on 21 July 2026 found no earlier public 22-variable cubic certificate, 44-variable quartic certificate, or explicit six-variable instantiation of the newly announced map. This is not an exhaustive claim of worldwide priority. The literature is changing by the hour and all novelty statements are provisional.

AI-assistance and verification disclosure

The constructions, searches, proofs, programs, website, and drafts were developed with heavy assistance from ChatGPT 5.6 Sol. Alec Kriebel is a complete amateur and cannot independently verify the claims. The note has not been peer reviewed; independent expert scrutiny is essential.

References

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- [8] W. Thompson, An explicit 24-variable cubic-homogeneous reduction of the Alpöge–Fable Jacobian counterexample, public GitHub repository, commit 45a7616fdf5a20c065564f2676190093722696b9, 21 July 2026; <https://github.com/wtho704/explicit-cubic-homogeneous-jacobian-counterexample>.
- [9] A. Kriebel, with heavy assistance from ChatGPT 5.6 Sol, An explicit Hessian-nilpotent quartic in 54 variables witnessing the failure of Zhao’s Vanishing Conjecture, provisional note, 21 July 2026.

Suggested citation

Alec Kriebel, with heavy assistance from ChatGPT 5.6 Sol, An explicit symmetric Keller counterexample in six variables and a 44-variable vanishing witness, provisional research note, first posted 21 July 2026, <https://aleckriebel.github.io/Math/papers/symmetric-keller-and-vanishing/>.